

DADIS: A Lane-Conditioned Risk-Aware Decision Framework for Vision-Based Driver Assistance

Abrar

Department of Electronics and Communication Engineering
Sri Venkateswara College of Engineering
Sriperumbudur, Tamil Nadu, India
Email: aabrar2703@gmail.com

Abstract

Advanced Driver Assistance Systems (ADAS) require reliable perception of surrounding traffic and the ability to convert perceptual information into timely and interpretable safety decisions. Vision-based systems provide a cost-effective sensing approach; however, independently detecting vehicles and lane markings does not directly determine which detected objects are relevant to the ego vehicle's immediate driving situation. This paper presents **DADIS (Driver Assistance and Decision Intelligence System)**, a modular vision-based driver assistance framework that integrates object detection, lane segmentation, ego-lane identification, monocular distance estimation, relative-motion analysis, time-to-collision estimation, collision-risk assessment, and lane-aware decision making. The proposed system assigns detected road participants to spatial lane regions and prioritizes the nearest relevant vehicle within the ego lane for longitudinal risk assessment. Vehicles outside the ego lane are retained as environmental context but do not independently trigger longitudinal braking or lane-change decisions. Based on estimated distance, relative motion, lane occupancy, and collision risk, the framework produces interpretable recommendations including **CLEAR, CAUTION, BRAKE, CHANGE LEFT, CHANGE RIGHT, and SLOW DOWN**. The system is implemented as a lightweight modular pipeline using deep-learning-based visual perception and conventional geometric and rule-based reasoning, allowing deployment on computationally constrained platforms. The framework is evaluated using driving video sequences under different traffic and environmental conditions. Evaluation considers object-detection performance, lane-estimation performance, computational latency, and decision consistency. The results are intended to demonstrate the feasibility of combining visual perception and lane-conditioned risk reasoning into an interpretable driver-assistance architecture.

Keywords — Advanced Driver Assistance Systems, computer vision, object detection, lane detection, lane segmentation, YOLO, collision risk, time-to-collision, lane-change decision, driver assistance.

I. INTRODUCTION

Road transportation environments contain continuously changing interactions between vehicles, lane boundaries, pedestrians, traffic signs, and other road users. For an intelligent driver assistance system, recognizing these objects is only the first stage of the problem. A practical system must determine which detected objects are relevant to the ego vehicle and translate the resulting scene representation into an appropriate safety recommendation.

Advanced Driver Assistance Systems have therefore evolved from isolated perception functions toward integrated systems that combine perception, scene understanding, risk estimation, and decision making. Camera-based approaches are particularly attractive because a monocular camera can provide information about vehicles, lane markings, road structure, traffic signs, and pedestrians without requiring a dedicated active ranging sensor.

Lane detection is one of the fundamental components of vision-based driving systems because lane geometry provides information about the position and direction of the ego vehicle relative to the road. Classical approaches have used image preprocessing, edge extraction, region-of-interest selection, and Hough transformation for identifying lane markings. The reference study provided for this work similarly investigates Canny edge detection, ROI selection, Hough transformation, CNN-based methods, and YOLO-based object recognition for road-scene analysis.

Deep-learning-based approaches have subsequently expanded lane perception capabilities. LaneNet, for example, formulates lane detection as a two-stage deep-learning problem involving lane-edge proposal and lane localization. Other approaches have formulated lanes as segmentation instances and attempted to overcome limitations associated with a fixed number of lanes.

Similarly, modern object detectors enable simultaneous identification of multiple road participants. YOLOv8 provides a real-time object-detection framework with an anchor-free detection head and support for multiple vision tasks. Driving datasets such as BDD100K provide large-scale and diverse road-scene data spanning different environments and weather conditions.

However, an important problem remains.

Consider a road scene containing three vehicles:



A conventional object-based risk system may consider all three vehicles when determining the closest or most dangerous object. This can produce inappropriate warnings when a nearby vehicle is travelling safely in an adjacent lane.

The central motivation of DADIS is therefore to establish a relationship between **object position and road-lane context** before making a driving decision.

The proposed system treats the ego lane as the primary region for longitudinal threat assessment. The nearest vehicle occupying the ego lane becomes the **front vehicle** used by the decision layer. Vehicles in adjacent lanes remain part of the world model and are evaluated when determining whether a lane change is feasible.

The main contributions of this work are:

1. A modular vision-based driver-assistance architecture combining object detection, lane segmentation, geometric reasoning, vehicle-state estimation, risk assessment, and decision making.
2. An **ego-lane-conditioned threat-selection mechanism** that distinguishes the vehicle directly relevant to the ego vehicle from surrounding vehicles travelling in adjacent lanes.
3. A **lane-aware decision mechanism** that evaluates left and right lane occupancy before recommending a lane-change manoeuvre.
4. An interpretable risk-assessment mechanism combining distance, lane position, relative motion, and time-to-collision information.
5. A lightweight implementation designed for real-time video processing on computationally constrained hardware.

Unlike a fully autonomous driving stack, DADIS is designed as a **driver-assistance and decision-support system**. It does not directly control steering, throttle, or braking actuators.

II. RELATED WORK

A. Vision-Based Lane Detection

Lane detection has traditionally relied on image-processing techniques. Canny edge detection is used to identify significant intensity transitions, after which a region of interest can restrict processing to the road region. Hough transformation can subsequently be applied to identify candidate lane lines. The reference paper used for this work describes this classical pipeline and notes the importance of ROI selection in reducing non-lane features.

Although computationally efficient, traditional approaches can become unreliable when lane markings are occluded, poorly illuminated, curved, faded, or affected by environmental

noise. The same reference work identifies noise and fog as factors that can reduce lane-detection performance.

Deep-learning methods address some of these limitations by learning lane representations directly from road-scene data. LaneNet uses lane-edge proposals followed by lane localization, providing a lane representation suitable for autonomous-driving applications. Instance-segmentation-based approaches have also been proposed to represent each lane as an independent instance and to support variable lane counts and lane changes.

DADIS uses learned lane segmentation as the perception component and subsequently applies geometric reasoning to derive an ego-lane representation used by the decision layer.

B. Object Detection in Driving Scenes

Object detection is required to identify vehicles and other road participants. Modern one-stage detectors provide a useful balance between accuracy and computational cost. YOLOv8 supports object detection and multiple related computer-vision tasks and uses an anchor-free detection architecture.

Large driving datasets have contributed significantly to the development of such perception systems. BDD100K contains 100,000 driving videos and provides multiple tasks for heterogeneous autonomous-driving research. Its geographic, environmental, and weather diversity makes it useful for evaluating perception systems under varied conditions.

The DADIS object detector is trained for road-scene classes including vehicles and other relevant traffic participants. Detection results provide the bounding box, class, and confidence required by subsequent modules.

C. Collision Risk and Decision Making

Time-to-collision (TTC) is widely used as a surrogate measure for identifying potentially hazardous interactions between vehicles. TTC can be expressed as the estimated time required for two approaching objects to collide under their current relative motion. IEEE literature has used TTC-based measures for identifying traffic conflicts and collision risk.

More advanced autonomous-driving systems incorporate uncertainty, vehicle interaction, trajectory prediction, and behaviour planning. For example, interaction-aware behaviour-planning research has incorporated collision risk and uncertainty into path and speed selection.

Such systems can provide sophisticated behaviour planning but may require considerably more computational and modelling complexity.

DADIS instead focuses on a lightweight and interpretable decision architecture in which perception outputs are transformed into a structured world model and subsequently into safety recommendations.

III. PROBLEM FORMULATION

Let the camera frame captured at time be represented by:

$$I_t$$

The object-detection network produces a set of detected objects:

$$O_t = \{o_1, o_2, \dots, o_n\}$$

Each detected object is represented as:

$$o_i = (c_i, b_i, p_i)$$

where

- c_i is the object class,
- $b_i = (x_1, y_1, x_2, y_2)$ is the bounding box,
- p_i is the detection confidence.

The lane segmentation module produces a lane representation:

$$L_t$$

From L_t , the system estimates lane geometry and derives a lane classification for each dynamic road participant:

$$\lambda_i \in \{\text{LEFT, CENTER, RIGHT, UNKNOWN}\}$$

For a vehicle V_i , its image-space center is calculated as:

$$x_i = (x_{1i} + x_{2i}) / 2$$

and its bottom-center point is:

$$y_i = x_{2i}^{\text{bottom}}$$

The estimated ego-lane center at the corresponding vertical position is represented by:

$$x_{\text{ego}}(y_i)$$

The lateral offset is:

$$\Delta x_i = x_i - x_{\text{ego}}(y_i)$$

Based on threshold values, the vehicle is assigned to the left, center, or right lane.

The primary front vehicle is then selected as:

$$V_f = \arg \min_{\{v_i \in V_{\text{CENTER}}\}} d_i$$

where represents vehicles classified as occupying the ego lane and is their estimated distance.

This formulation ensures that a vehicle in an adjacent lane does not automatically become the primary threat simply because it is visually closer.

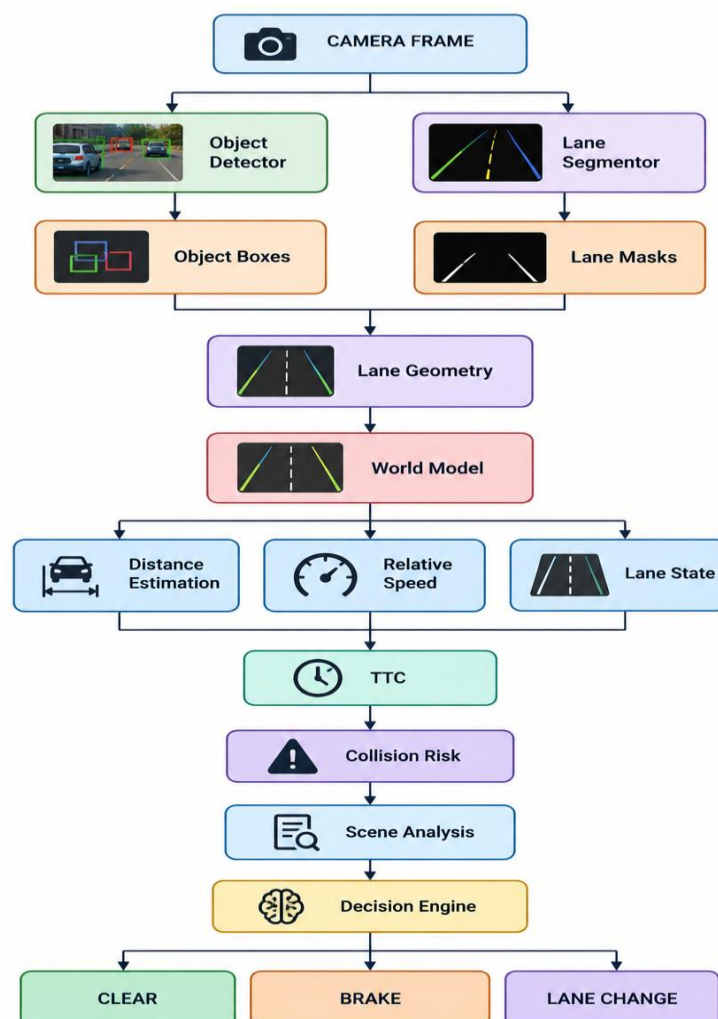
IV. PROPOSED DADIS ARCHITECTURE

A. System Overview

The proposed architecture is divided into six major layers:

1. Visual perception
2. Lane geometry
3. World modelling
4. Vehicle-state estimation
5. Risk assessment
6. Decision generation

The complete pipeline is:



B. Object Detection

DADIS uses a YOLO-based detector to identify road-scene objects.

The trained detector contains ten classes:

ID Class

- 0 Bus
- 1 Traffic Light
- 2 Traffic Sign
- 3 Person
- 4 Bike
- 5 Truck
- 6 Motor
- 7 Car
- 8 Train
- 9 Rider

The detector produces:

$$B_i=(x1,y1,x2,y2,ci,pi)$$

for each object.

A confidence threshold is applied to suppress low-confidence detections.

For the risk-assessment pipeline, dynamic road participants are distinguished from static/contextual classes.

Dynamic classes include:

{car, truck, bus, motor, bike, person, rider, train}

Traffic signs and traffic lights are detected and displayed but are not assigned vehicle-state quantities such as TTC or relative speed.

C. Lane Segmentation

The lane segmentation module produces a pixel-level representation of lane markings.

The segmentation output is converted into a binary lane mask:

$$M(x, y) = 1, \text{lanepixel}; 0, \text{otherwise}$$

The mask is resized to the original video-frame dimensions.

The current implementation extracts lane geometry from the segmentation output rather than relying exclusively on classical edge detection.

D. Ego-Lane Estimation

For each horizontal image row containing valid lane pixels, the leftmost and rightmost lane pixels are identified.

If:

$$x_L(y)$$

And

$$x_R(y)$$

represent the detected lane boundaries, the lane center is estimated as:

$$x_C(y) = \frac{x_L(y) + x_R(y)}{2}$$

A sequence of center points:

$$P = \{(x_C(y_1), y_1), \dots, (x_C(y_n), y_n)\}$$

is generated.

For a detected vehicle, the center of its bounding box is compared with the closest lane-center point at approximately the same vertical position.

The lateral displacement is:

$$\Delta x = x_v - x_C$$

The vehicle is classified according to:

$$\begin{aligned} \text{Lane}(v) = & \\ & \{ \\ & \text{LEFT}, \&\Delta x < -T \\ & \text{CENTER}, \&|\Delta x| \leq T \\ & \text{RIGHT}, \&\Delta x > T \end{aligned}$$

where T is the lateral classification threshold.

This lane classification forms the basis of the ego-lane-conditioned threat-selection mechanism.

E. World Model

Instead of passing independent detections between modules, DADIS creates a structured representation for each detected road participant.

Each vehicle contains:

$$V_i = (id, class, bbox, confidence, distance, relative\ speed, TTC, lane, risk)$$

The world model is cleared and reconstructed for each video frame.

This provides a common representation that can be accessed by the risk and decision modules.

F. Monocular Distance Estimation

The system estimates distance using the pinhole-camera relationship:

$$d = \frac{Hf}{h}$$

where:

- d = estimated distance,
- H = assumed real-world object height,
- f = camera focal length,
- h = observed pixel height.

The implementation uses class-dependent approximate object heights:

Object	Assumed height
Car	1.5 m
Bus	3.2 m
Truck	3.5 m

For classes without a specific height model, a default height is used.

This approach is computationally inexpensive and requires no additional depth sensor. However, the resulting distance should be considered an estimate rather than a calibrated measurement.

G. Relative Motion and TTC

Relative motion is estimated from temporal changes in the observed vehicle state.

The system classifies the vehicle into states such as:

- **CLOSING FAST**
- **CLOSING**
- **MOVING AWAY**
- **UNKNOWN**

The time-to-collision is then estimated when the relative motion indicates an approaching vehicle.

Conceptually:

$$TTC = \frac{d}{v_{rel}}$$

where d represents estimated separation and v_{rel} represents closing velocity.

For:

$$v_{rel} \leq 0$$

TTC is considered non-threatening or undefined.

TTC is used as one component of the overall risk calculation rather than as the sole decision variable.

H. Collision-Risk Assessment

The system combines multiple indicators into a risk score.

The current risk formulation is:

$$R = 0.25D + 0.15L + 0.10V + 0.50C$$

where:

- D = distance-risk score,
- L = lane-position score,
- V = vehicle-type score,
- C = collision-risk component.

The distance-risk component increases as the estimated vehicle distance decreases.

The lane-risk component gives the highest weighting to the ego lane:

$$L = \begin{cases} 1.0, & \text{CENTER} \\ 0.4, & \text{LEFT/RIGHT} \\ 0.1, & \text{UNKNOWN} \end{cases}$$

This is an important component of the proposed architecture because the same physical distance should not necessarily correspond to the same threat level for vehicles occupying different lanes.

I. Lane-Aware Scene Analysis

The scene analyzer separates detected vehicles into:

$$V_{left}, \quad V_{center}, \quad V_{right}$$

The nearest vehicle in each lane is retained.

The front vehicle is:

$$v_f = \arg \min_{v \in V_{center}} d(v)$$

The left and right lanes are considered occupied when a vehicle is detected within a predefined safety distance.

This results in:

$$S = \{v_f, v_{left}, v_{right}, L_{left}, L_{right}\}$$

where L_{left} and L_{right} indicate whether the corresponding adjacent lane is considered sufficiently clear.

J. Decision Engine

The decision engine converts scene information into an interpretable recommendation.

If no vehicle exists in the ego lane:

$$v_f = \emptyset$$

the system returns:

CLEAR — Lane Clear

No braking or lane-change recommendation is generated solely because of vehicles travelling in adjacent lanes.

If a front vehicle exists and the collision risk becomes severe, the system generates:

BRAKE

For a high-risk front vehicle, the adjacent lanes are evaluated.

If the left lane is safe:

CHANGE LEFT

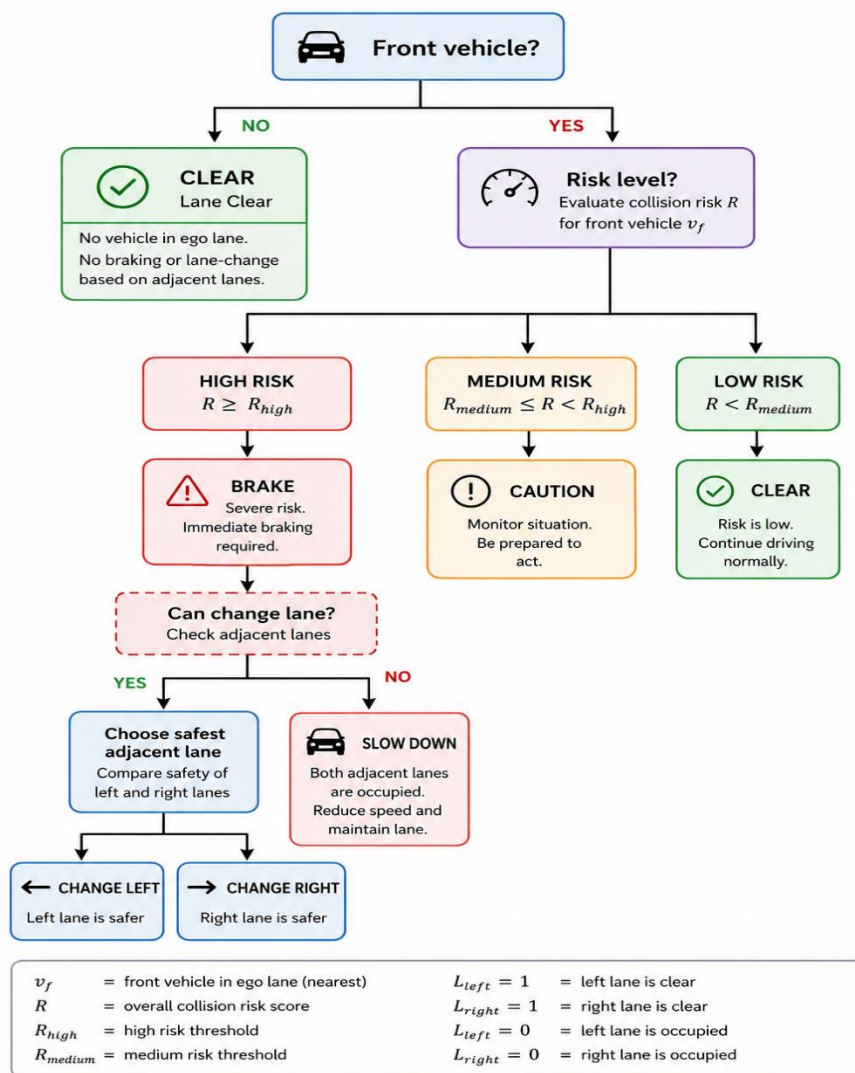
If the left lane is unavailable but the right lane is safe:

CHANGE RIGHT

If both adjacent lanes are occupied:

SLOW DOWN

The conceptual decision structure is:



V. IMPLEMENTATION

The system is implemented in Python using:

- Python 3.x
- OpenCV
- NumPy

- PyTorch
- Ultralytics
- YOLO
- custom Python modules for risk and decision reasoning

The software is divided into independent modules to simplify development and testing.

The principal modules are:

- object_detection.py
- lane_segmentation.py
- lane_geometry.py
- lane_classifier.py
- world_model.py
- world_analyzer.py
- distance_estimator.py
- relative_speed.py
- ttc.py
- motion_state.py
- collision_probability.py
- risk_assessment.py
- decision_engine.py
- main_pipeline.py

This modular architecture allows the perception models to be replaced independently of the decision layer.

VI. EXPERIMENTAL SETUP

A. Dataset

The object-detection component was trained using a road-scene dataset configured in YOLO format.

The dataset configuration contains ten classes corresponding to road users and traffic-related objects.

The lane perception component was trained using the **CULane** dataset and a YOLO-compatible lane-segmentation representation.

The CULane dataset is particularly relevant because lane detection must operate under diverse road configurations and challenging visual conditions. Modern lane-detection research has also emphasized the difficulties created by lane markings, lane-count variation, road appearance, and environmental conditions.

B. Training Configuration

The object detector was trained using:

- Model: YOLO-based detector
- Image size: 640×640
- Epochs: 100
- Batch size: 16
- Optimizer: AdamW
- Training images: 8,000
- Validation images: 2,000

The lane segmentation model was trained using:

- Dataset: CULane
- Model: YOLOv8n-Seg
- Image size: 640×640
- Epochs: 40
- Batch size: 16

VII. RESULTS AND DISCUSSION

A. Object Detection Performance

The object detector should be evaluated using:

- Precision
- Recall
- mAP@50
- mAP@50–95

TABLE I

OBJECT DETECTION PERFORMANCE

Metric	Value
Precision	
Recall	
mAP@50	
mAP@50–95	

Your earlier validation run should **not** automatically be used here because the detector/model configuration changed during development. The final model needs one clean validation run.

B. Lane Detection Performance

TABLE II

LANE DETECTION PERFORMANCE

Metric	Value
IoU	
Dice/F1	
Lane detection accuracy	
Average inference time	

C. System Performance

The complete DADIS pipeline is evaluated according to computational latency.

TABLE III

SYSTEM PERFORMANCE

Parameter	Value
Input resolution	640 × 384
Average inference time	~493–547 ms/frame
Average FPS	~2 FPS
CPU/GPU	CPU — Intel Core i5-1235U / Iris Xe system
RAM	8GB
Object detector	YOLO
Lane detector	YOLOv8n-Seg

The current implementation demonstrates that combining perception and decision modules is feasible on consumer-grade hardware. However, the computational cost of running multiple neural networks sequentially represents a significant limitation for high-frame-rate deployment.

VIII. DECISION-MAKING EVALUATION

To evaluate the lane-conditioned decision mechanism, driving scenarios are categorized according to the location and behavior of surrounding vehicles.

TABLE IV

DECISION SCENARIOS

Scenario	Expected Output
No vehicle in ego lane	CLEAR
Safe vehicle ahead	CLEAR
Vehicle approaching ego vehicle	CAUTION
High-risk vehicle ahead	BRAKE
High-risk vehicle + left lane free	CHANGE LEFT
High-risk vehicle + right lane free	CHANGE RIGHT
High-risk vehicle + both lanes occupied	SLOW DOWN
Vehicle only in adjacent lane	No longitudinal warning

The final paper should report the number of correctly classified scenarios:

$$Accuracy_{decision} = \frac{N_{correct}}{N_{total}} \times 100$$

IX. ABLATION STUDY

A key experiment for evaluating the contribution of the proposed lane-conditioned reasoning is to compare different system configurations.

TABLE V

ABLATION OF DADIS COMPONENTS

Configuration	Ego-Lane Awareness	TTC Risk Model	Decision Accuracy
Object-only	No	No	No
Object + TTC	No	Yes	Partial
Object + Lane	Yes	No	Partial
Object + Lane + TTC	Yes	Yes	Yes
DADIS	Yes	Yes	Yes

The purpose of this experiment is to determine whether lane-conditioned reasoning provides measurable benefits over object-only risk assessment.

In particular, the system should be tested on situations in which an adjacent-lane vehicle is closer to the camera than the vehicle in the ego lane.

The expected advantage of DADIS is that the adjacent-lane vehicle should not independently trigger a longitudinal emergency decision.

X. DISCUSSION

The proposed framework demonstrates the importance of connecting object perception with lane context.

A purely object-centric system may observe:

Vehicle A = 25 m

Vehicle B = 32 m

Vehicle C = 40 m

and select Vehicle A as the most relevant object.

DADIS instead introduces spatial context:

Vehicle A → LEFT

Vehicle B → CENTER

Vehicle C → RIGHT

and therefore selects:

Vehicle_B

as the primary longitudinal threat.

This distinction is particularly important for multi-lane roads where vehicles in adjacent lanes can be physically closer to the camera than the vehicle directly ahead.

The proposed architecture therefore separates:

1. **Object detection**
2. **Spatial lane association**
3. **Threat selection**
4. **Risk estimation**
5. **Decision generation**

This modular decomposition also improves interpretability because every decision can be traced to measurable quantities such as lane position, distance, relative motion, and TTC.

XI. LIMITATIONS

Despite its modular architecture, DADIS has several limitations.

1. Monocular distance estimation

The distance estimator relies on assumed object dimensions and a fixed focal-length approximation. Therefore, estimated distances may vary with camera calibration, vehicle type, camera mounting position, and perspective.

2. Lane perception dependency

The ego-lane classifier depends strongly on the quality of lane segmentation. Poor or missing lane markings can cause incorrect lane assignment.

3. Relative-speed estimation

Relative speed is inferred from temporal visual observations rather than measured directly using radar or other sensors. Consequently, the estimate is sensitive to detection noise and camera motion.

4. Rule-based decisions

The current decision engine is deterministic and rule-based. It does not learn driver behavior or predict long-term trajectories.

5. Collision-risk score

The current collision-risk value is a heuristic score and should not be interpreted as a calibrated probability of collision unless it is trained and validated against appropriate ground-truth collision data.

6. No actuator control

DADIS provides driver-assistance recommendations and does not directly control the vehicle's steering, throttle, or braking systems.

XII. FUTURE WORK

Several extensions can improve the system.

First, temporal filtering can be introduced to stabilize ego-lane estimation across consecutive frames. This would reduce sudden lane-classification changes caused by individual segmentation errors.

Second, the current heuristic risk score can be replaced with a learned or probabilistically calibrated risk model trained on labeled driving scenarios.

Third, monocular depth estimation or stereo/radar fusion can improve distance accuracy.

Fourth, vehicle tracking can be incorporated to maintain consistent vehicle identities across frames.

Finally, the decision layer can be extended from discrete recommendations toward trajectory-level planning in which candidate maneuvers are evaluated according to collision risk, lane availability, vehicle dynamics, and driver comfort.

XIII. CONCLUSION

This paper presented DADIS, a modular vision-based driver assistance and decision-intelligence framework integrating object detection, lane segmentation, ego-lane identification, distance estimation, relative-motion analysis, time-to-collision estimation, collision-risk assessment, and lane-aware decision making.

The central design principle of DADIS is that **object proximity alone is insufficient for determining driving risk**. The relevance of a vehicle depends strongly on its relationship to the ego vehicle's lane. By associating detected vehicles with lane regions and selecting the nearest vehicle in the ego lane as the primary longitudinal threat, the system provides a more context-aware representation of the driving scene.

The framework further evaluates adjacent lane occupancy before recommending a lane-change maneuver and provides interpretable outputs such as CLEAR, CAUTION, BRAKE, CHANGE LEFT, CHANGE RIGHT, and SLOW DOWN.

The system is intentionally designed as a lightweight driver-assistance framework rather than a complete autonomous-driving stack. Its modular architecture allows individual perception components to be improved independently while retaining the same world-model and decision interfaces.

Future work will focus on temporal ego-lane estimation, calibrated collision-risk prediction, improved monocular depth estimation, multi-object tracking, and uncertainty-aware maneuver selection.

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